



A deep spatio-temporal Graph Attention Network to learn nonlinear operators for traffic prediction

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ABSTRACT

Accurate long-horizon traffic prediction is critical for intelligent transportation systems, yet it remains challenging due to the highly nonlinear spatio-temporal dependencies and complex temporal evolution of traffic networks. Existing sequence-to-sequence prediction models often suffer from error accumulation and degraded accuracy in long-horizon forecasts. To address these challenges, we propose the Deep Spatial-Temporal Graph Attention Operator Network (DSTGAON), which integrates operator learning with graph attention mechanisms for traffic speed prediction. Inspired by the Mori–Zwanzig formalism from statistical mechanics, DSTGAON models long-horizon traffic evolution from the perspective of nonlinear operator approximation and non-Markovian memory effects. Specifically, the framework extracts key spatio-temporal features from historical traffic states using a branch network and encodes future temporal query information via a trunk network; the outputs of the two networks are fused through a dot product to predict traffic state over specified future horizons. This design provides a structured approach to capture dynamic spatial correlations, temporal dependencies, and nonlinear traffic evolution, thereby significantly enhancing long-horizon prediction performance. Extensive experiments on benchmark traffic datasets demonstrate that DSTGAON outperforms state-of-the-art baselines, particularly in long-horizon forecasting, and additional fitting and stability analyses further confirm its reliability under diverse traffic conditions.

1. Introduction

Highway traffic speed prediction is a fundamental task in intelligent transportation systems (ITS). It supports proactive traffic control by providing short- and medium-term information on future traffic states [1]. It also provides a basis for route guidance, travel time estimation, and traffic management decisions, thereby helping transportation agencies alleviate congestion and improve network operations [2,3]. Recent surveys have further emphasized that accurate traffic prediction is becoming increasingly important for data-driven ITS applications [4]. However, accurate long-horizon prediction remains challenging because highway traffic evolves as a nonlinear spatio-temporal dynamical process. Traffic states are affected not only by local temporal fluctuations, but also by spatial interactions among connected road segments and by memory effects induced by previous congestion states. As the forecasting horizon increases, prediction errors may accumulate over time and propagate across the road network, resulting in degraded accuracy and unstable performance under different traffic conditions.

In the long-term effort to overcome the above difficulties, two paradigms have gradually emerged: model-driven and data-driven approaches. Model-driven methods [5–7] rely on traffic flow theory and prior knowledge to construct physical descriptions of traffic dynamics. While these methods theoretically capture fundamental characteristics of traffic flow, their dependence on

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idealized assumptions limits their predictive accuracy in complex real-world scenarios. Data-driven approaches [4,8,9] have been developed to compensate for these limitations by learning patterns directly from observational data, demonstrating superior empirical performance without requiring explicit physical formulations. However, data-driven methods still face several fundamental challenges. First, learning direct mappings between input and output functions provides no principled mechanism to control error accumulation, particularly in long-horizon forecasting. Second, spatial and temporal dependencies are often modeled in isolation or through heuristic fusion, lacking a unified mathematical framework for their coupled dynamics. Third, the deep learning paradigm is largely perceived as a “black box”, offering limited interpretability. These challenges collectively motivate the need for a theoretically grounded framework that can capture nonlinear spatio-temporal dynamics while suppressing error propagation and providing interpretable representations.

Recently, operator learning has emerged as a promising paradigm that generalizes traditional deep learning from learning finite-dimensional mappings to approximating nonlinear operators between infinite-dimensional function spaces [10]. Unlike conventional methods, operator learning directly learns the underlying solution operator of a dynamic system, providing a theoretically grounded framework that is inherently interpretable and resistant to error accumulation. Building upon the Mori–Zwanzig formalism from statistical mechanics [11], which provides a rigorous mathematical framework for decomposing large-scale systems and modeling non-Markovian memory effects, operator learning offers a principled pathway to address the aforementioned challenges in traffic prediction.

In this paper, we propose the Deep Spatial-Temporal Graph Attention Operator Network (DSTGAON), a spatio-temporal graph operator learning framework for long-horizon highway traffic speed prediction. DSTGAON integrates graph attention mechanisms with a branch–trunk operator architecture. Specifically, the branch network is constructed using stacked spatio-temporal blocks (ST-Blocks) to encode historical traffic states and capture dynamic spatial–temporal correlations on irregular road networks, while the trunk network encodes future temporal query information. The two representations are then fused to approximate nonlinear traffic evolution operators and generate predictions at flexible future horizons. The main contributions of this work are summarized as follows:

1. Grounded in the Mori–Zwanzig formalism, we decompose large-scale traffic networks into subsystems with history-dependent terms, and propose a branch–trunk spatio-temporal graph operator approximation method. This approach performs key feature extraction from historical traffic states and maps them to future predictions, effectively approximating the nonlinear solution operator while suppressing error accumulation in long-horizon forecasting.
2. Within the branch network, we embed spatio-temporal blocks that combine graph attention with gated temporal convolution, where graph attention captures dynamic dependencies among traffic network nodes and gating mechanisms adaptively filter key features across historical time steps, enhancing the utilization of historical information. Combined with the trunk network that encodes future temporal queries, this design achieves efficient fusion of historical features and future temporal queries, improving both the stability and accuracy of long-horizon predictions.
3. On multiple datasets, DSTGAON demonstrates superior long-horizon prediction accuracy, stability, and reliability. Moreover, the proposed operator framework can serve as a plug-and-play module that, when embedded into other traffic forecasting models, yields consistent performance gains without altering their original architectures or hyperparameters, validating its generalization capability and practical value across diverse models and scenarios.

The organization of the paper is as follows: Section 2 reviews related studies on traffic prediction, Transformer-based forecasting, and operator learning. Section 3 introduces the preliminaries, including notations and operator theory. Section 4 presents the proposed methodology in detail. Section 5 describes the experimental datasets, baselines, experimental results, and discussions. The main conclusions and future prospects are summarized in Section 6.

2. Related work

2.1. Traditional methods and deep learning approaches

Early traffic prediction research relied on statistical methods such as ARIMA and SARIMA, which capture autocorrelation and seasonal patterns but assume stationarity and linearity, limiting their ability to model complex nonlinear traffic dynamics [8,12,13]. Traditional machine learning methods, including Support Vector Regression (SVR) and Bayesian Networks, improved nonlinear learning capability but still only capture shallow spatio-temporal features [9,14]. Model-driven approaches, such as the Lighthill–Whitham–Richards (LWR), Payne–Whitham (PW), and Aw–Rascle–Zhang (ARZ) models, construct physical descriptions from traffic flow theory but rely on idealized assumptions that limit real-world accuracy [5–7].

The advent of deep learning enabled more powerful feature extraction. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks capture nonlinear temporal variations through gating mechanisms, yet largely overlook spatial dependencies [15–17]. CNN-based methods such as ST-ResNet incorporate spatial context but are inherently designed for regular grids, making them ill-suited for irregular road network topologies [18]. To address this, Graph Neural Networks (GNNs) have become the dominant paradigm. DCRNN combines diffusion convolution with recurrent networks for spatio-temporal graph modeling, while AGCRN introduces adaptive graph convolution that dynamically adjusts to input data [19,20]. More recently, integrated approaches have sought to jointly model spatial and temporal features: Yang et al. [21] employed recalling attention to alleviate error accumulation, and Xu et al. [22] proposed heterogeneous spatio-temporal graphs for asynchronous decoupling. Despite these advances, GNN-based methods often rely on precise graph construction and remain largely “black box” in nature, limiting their interpretability and robustness.

2.2. Attention mechanisms and transformer-based spatio-temporal forecasting

Attention mechanisms have significantly advanced traffic prediction by enabling adaptive weighting of spatial and temporal dependencies. The Graph Attention Network (GAT) allows neighbor weighting without a fixed adjacency matrix, making it suitable for inductive learning on graph-structured data [23]. ASTGCN and STGAT further integrate spatial and temporal attention to dynamically capture evolving correlations [24,25]. Building on this, Transformer architectures have been widely adopted for traffic and time series forecasting. TrafficBERT pioneered pre-trained Transformer models with multi-head self-attention for traffic flow [26]. iTransformer inverted the conventional design by embedding entire variate time series as tokens, mitigating performance degradation with large lookback windows [27]. For spatio-temporal data, STAEformer achieves state-of-the-art results via adaptive embedding with a vanilla Transformer [28], while Fredformer addresses spectral bias through frequency debiasing [29]. Zou et al. [30] validated iTransformer for freeway speed prediction, showing particular strength beyond 60-minute horizons.

Leveraging Transformer advances, large language models (LLMs) have recently been explored for spatio-temporal forecasting. STG-LLM bridges LLMs to traffic data via an STG-Tokenizer that encodes graph-structured relationships into tokens and a lightweight STG-Adapter, achieving competitive performance through parameter-efficient fine-tuning [31]. UniST adopts a universal pre-training paradigm across diverse spatio-temporal scenarios with knowledge-guided prompts, demonstrating strong few-shot and zero-shot generalization [32]. Although Transformer-based traffic prediction methods demonstrate strong empirical performance, they typically formulate the prediction task as a discrete sequence modeling problem, with the connection to the continuous nonlinear evolution operators in traffic dynamics often being implicit. Moreover, both Transformers and large language model-based approaches face inherent limitations, including the quadratic computational complexity of self-attention mechanisms and the modality gap between discrete tokens and continuous dynamics.

2.3. Operator learning for spatio-temporal prediction

Operator learning represents a paradigm shift from learning finite-dimensional mappings to approximating nonlinear operators between function spaces. DeepONet [10] provides a universal approximation theorem for nonlinear operators via a branch-trunk architecture, and has been successfully applied to fluid dynamics, power grid trajectory prediction, multiphase flow, and nuclear energy applications [33–36]. Sun et al. [37] extended this framework to graph-structured networked systems with DeepGraphONet, fusing GNNs with DeepONet for resolution-independent prediction and zero-shot transfer. In the traffic domain, Yu et al. [38] applied DeepONet-based branch-trunk factorization for speed forecasting with cross-scenario generalization, while Li et al. [39] proposed PI-DeepONet, embedding traffic flow conservation laws directly into the operator learning process for physically consistent predictions. By directly learning the solution operator of dynamical systems, operator learning inherently addresses the black-box limitation, suppresses error accumulation, and enables evaluation at arbitrary time horizons. This framework for spatio-temporal prediction motivates the DSTGAON architecture proposed in this work.

3. Preliminaries

3.1. Notations

The notations and symbols used in this study are listed in Table 1.

3.2. Complex system prediction

Consider a complex traffic network system, as shown in Eq. (1), which describes the continuous evolution of traffic states such as speed and flow across the road network graph G . We represent the connections between sensors in the traffic network as a weighted undirected graph $G = (\mathcal{V}, E, A)$, where $\mathcal{V}$ denotes the set of nodes, and $|\mathcal{V}| = N$. E is the set of edges, and $A \in \mathbb{R}^{N \times N}$ represents the weighted adjacency matrix of graph G .

$$\begin{cases} \frac{d}{dt}u(t) = f(u(t); G), \\ u(t_0) = u_0, \end{cases} \quad (1)$$

where $u(t) \in X \subset \mathbb{R}^{d \times N}$ represents the traffic state function observed on the graph at time t . Here, $u(t_0)$ denotes the initial traffic state at the initial time t_0 , u_0 is the corresponding initial state value. Given the graph G , the traffic flow prediction problem becomes the process of learning the operator that maps historical traffic states to future traffic responses.

Table 1

List of notations.

Symbol	Definition
G	Highway network represented as a weighted undirected graph.
$\mathcal{V}, v_i$	Set of nodes, i.e., traffic sensors; node i .
E	Set of edges, i.e., road links.
A	Weighted adjacency matrix of graph G .
N	Number of nodes.
F	Solution operator mapping historical states to future states.
t, t_0	Current time step; initial observation time.
k	Number of historical input time steps.
p	Prediction horizon, i.e., number of future time steps.
d	Number of features at each node, e.g., speed, flow, and occupancy.
$u = \{u_{t-k+1}, \dots, u_t\}$	Historical traffic observations.
$\hat{u} = \{\hat{u}_{t+1}, \dots, \hat{u}_{t+p}\}$	Predicted traffic states.
$u(t)$	The traffic flow state function observed on the graph at time t .
S, S^c	The sub-network region; the complement of S .
κ	Error bound of the truncated memory approximation.
$A(u_S(t); S)$	Markov state term of subsystem S .
$K(\cdot)$	Memory kernel function in the Mori–Zwanzig formalism.
$Q(\cdot)$	Discrete approximation of the memory integral using historical states.
τ_i, m, t_M	Historical lag time index; number of selected lag terms; maximum memory length.
h, h_n	Prediction horizon length; local prediction step size.
r	Latent basis dimension used in the branch–trunk fusion.
$dsc_{(v_i, t)}^l$	Dynamic spatial correlation of node v_i at time t in layer l
$dsc_{(v_i, t)}^{(l, m)}$	Dynamic spatial correlation of node v_i at time t in layer l , head m
$dsc_{(v_i, t)}$	Final dynamic spatial correlation of v_i at time t after multi-head aggregation.
$dte(X)$	Dynamic temporal correlation extracted from input sequence X .
$dste$	Dynamic spatio-temporal correlation after fusing spatial and temporal correlations.
$\sigma_q^m, \sigma_k^m, \sigma_v^m$	Nonlinear transformation functions for query, key, and value in head m .
$c_{(v_i, v_j)}^{(l, m)}$	Correlation score between node v_i and node v_j at layer l , head m .
$\alpha_{(v_i, v_j)}^{(l, m)}$	Spatial attention coefficient between node v_i and node v_j at time t , layer l , head m .

3.3. Operator solution

To approximate the dynamic system described by the complex traffic network system Eq. (1), we have the following Eq. (2):

$$F(u(t_0); G)(t) \equiv u(t) = u(t_0) + \int_{t_0}^t f(u(\beta); G) d\beta, \quad (2)$$

For all t , F denotes the solution operator, which exists and is unique for a finite time interval. The operator F maps the initial traffic state $u(t_0)$ to the future state $u(t)$ by integrating the network dynamics.

However, when $N = |\mathcal{V}| \gg 1$, directly learning such a large-scale dynamic traffic network system consumes significant computational resources. Therefore, we decompose the large-scale network into several smaller sub-networks, representing the graph-valued state function $u(t)$ as a composition $u = (u_S(t), u_{S^c}(t))$, where $S = (\mathcal{V}_S, E_S)$ denotes sub-network regions. Here, $\mathcal{V}_S \subset \mathcal{V}$ and $E_S \subset E$ respectively represent the sets of nodes and edges in the sub-networks, and S^c denotes the complement of S .

To describe the dynamic traffic system of the sub-networks, we use the Mori–Zwanzig formalism equation [11], which incorporates the effects of unresolved dynamics through the memory kernel and orthogonal dynamics:

$$\frac{d}{dt} u_S(t) = A(u_S(t); S) + \int_{t_0}^t K(u_S(t - \beta), \beta) d\beta + B(u_0), \quad (3)$$

The first term $A(u_S(t); S)$ is the Markov state term, which describes the instantaneous evolution of the resolved sub-network state $u_S(t)$ on sub-network S . The second term is the memory integral, which accounts for the influence of historical states from the initial time $\beta = t_0$ to the current time $\beta = t$ through the memory kernel K . The third term $B(u_0)$ is the orthogonal dynamics or random force term determined by the initial traffic state u_0 , representing the unresolved influence of the complement sub-network S^c on the resolved state $u_S(t)$.

At this point, we have transformed the original problem into a more manageable one by focusing on $u_S(t)$ and incorporating the rest of the network's influence through the memory term and the unresolved contribution $B(u_0)$.

3.4. Discrete representation

The integral $\int_{t_0}^t K(u_S(t - \beta), \beta) d\beta$ is computationally intensive, especially for long time intervals. To obtain an approximate solution for the dynamic system $u_S(t)$, we apply a discretization method to the memory integral term in Eq. (3) to compute the value of the

truncated integral. By discretizing the truncated integral with a prescribed approximation tolerance $\kappa > 0$, the integral term in Eq. (3) is represented as follows:

$$\left| \int_{t_0}^{t^M} K(u_S(t - \beta), \beta) d\beta - Q(u_S(t - \tau_1), \dots, u_S(t - \tau_m)) \right| < \kappa, \quad (4)$$

The parameter $m \in [0, M]$ satisfies $0 \leq \tau_1, \dots, \tau_m \leq t_M$ and $\{t - \tau_i\}_{i=1}^m \subset [t - t_M, t]$. Here $Q(u_S(t - \tau_1), \dots, u_S(t - \tau_m))$ is a function approximating the integral using discrete past states, where τ_i denotes the i th time lag and t_M denotes the maximum memory length considered in the truncated history.

Combining Eq. (3) and Eq. (4), we obtain an approximate representation of the dynamic system $u_S(t)$:

$$\frac{d}{dt} u_S(t) = A(u_S(t); S) + Q(u_S(t - \tau_1), \dots, u_S(t - \tau_m)) \quad (5)$$

In this approximation, the unresolved contribution associated with $B(u_0)$ is not modeled as an independent forcing term. Instead, its effect is absorbed into the data-driven memory approximation $Q(\cdot)$, which uses the discrete historical states to represent the combined influence of memory and unresolved dynamics.

Furthermore, let $h_n \in [0, h]$. Within any time interval $[t, t + h_n]$, we can use a local solution operator to solve the dynamic system. This operator predicts the traffic state h_n steps ahead using both current and historical states. For all $t > t_0$, with the local solution operator's time query vector satisfying $\{t - \tau_i\}_{i=1}^m \subset [t - t_M, t]$, we can obtain the solution of the local solution operator as follows:

$$\tilde{F}(u_S(t), \{u_S(t - \tau_i)\}_{i=1}^m; S)(h_n) = u_S(t) + \int_t^{t+h_n} A(u_S(\beta); S) d\beta + Q(u_S(t - \tau_1), \dots, u_S(t - \tau_m)). \quad (6)$$

3.5. Operator approximation theory

Define σ as a continuous non-polynomial function, $K_1 \subset X$, $K_2 \subset \mathbb{R}^d$ are two compact sets in X and $\mathbb{R}^d$, respectively. Let $\mathcal{U}$ be a compact set in $C(K_1)$, and F be a nonlinear continuous operator. For any $\epsilon > 0$, there exist positive integers n, p, m and constants $c_i^k, \xi_{ij}^k, \theta_i^k, w_k$, and ζ_k such that Eq. (7) holds [10]:

$$\left| F(u)(y) - \sum_{k=1}^p \sum_{i=1}^n c_i^k \underbrace{\sigma\left(\sum_{j=1}^m \xi_{ij}^k u(x_j) + \theta_i^k\right)}_{\text{branch net}} \underbrace{\sigma(w_k \cdot y + \zeta_k)}_{\text{trunk net}} \right| < \epsilon, \quad \forall u \in \mathcal{U}, y \in K_2. \quad (7)$$

In Eq. (7), u denotes an input function sampled at sensor locations or historical observation points $\{x_j\}_{j=1}^m$, y denotes the query variable at which the operator output is evaluated, c_i^k are output coefficients, ξ_{ij}^k and w_k are learnable weights, and θ_i^k and ζ_k are bias terms. The integers m, n , and p denote the number of input sampling points, branch-network basis terms, and trunk-network basis terms, respectively.

By combining Eq. (6) and Eq. (7), we show that neural operators can approximate the solution operator of traffic dynamics. This captures both spatial dependencies (via graph structure G) and temporal memory (via operator formalism), providing a data-driven approximation of the dynamic operator F and enabling long-horizon traffic prediction. Therefore, the operator approximation theorem provides a theoretical basis for using a branch-trunk neural architecture to learn nonlinear spatio-temporal traffic dynamics from sensor observations.

4. Methodology

4.1. Problem formulation

The objective of this study is to predict traffic speed at a specific future moment based on historical observed data. To incorporate temporal and spatial influences, data collected by detectors at the target location and its surrounding areas are used as inputs. We represent the connections between sensors in the highway as a weighted undirected graph $G = (\mathcal{V}, E, A)$, where $\mathcal{V}$ denotes the set of nodes, and $|\mathcal{V}| = N$. E is the set of edges, and $A \in \mathbb{R}^{N \times N}$ represents the weighted adjacency matrix of graph G .

Given k input time steps and p prediction time steps, for N nodes, the traffic detector speed data for the first k time stamps $u = \{u_{t-k+1}, u_{t-k+2}, \dots, u_t\} \in \mathbb{R}^{k \times N \times d}$ are used to predict the speed data for the future p time stamps $\hat{u} = \{\hat{u}_{t+1}, \dots, \hat{u}_{t+p}\} \in \mathbb{R}^{p \times N \times d}$, where d denotes the number of traffic features at each node.

$$\{u_{t-k+1}, u_{t-k+2}, \dots, u_t; G\} \longrightarrow \{\hat{u}_{t+1}, \dots, \hat{u}_{t+p}; G\}, \quad (8)$$

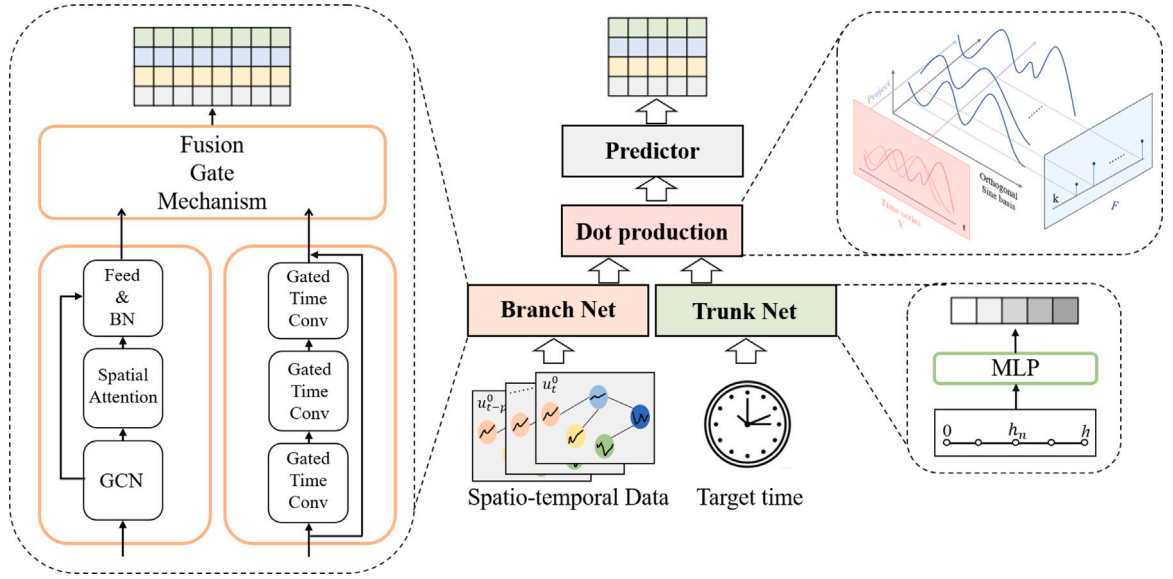


Fig. 1. The framework of the proposed DSTGAON.

4.2. Framework of DSTGAON

Fig. 1 illustrates the proposed model DSTGAON, designed to predict traffic speed for various segments of the highway traffic networks. The network comprises a Branch network and a Trunk network. The outputs of the Branch and Trunk networks represent basis function values of the target function and the coefficients, respectively. The Trunk network uses a multilayer perceptron (MLP) architecture to generate trainable coefficients that depend only on the future temporal query. In the Branch network, we designed spatio-temporal blocks (ST-Blocks) to extract complex temporal and spatial correlations from the traffic data, the details of which will be discussed in the next section. Finally, the basis function values b and basis function coefficients y produced by the Branch and Trunk networks are combined through a dot product, as shown in Eq. (9), to generate the final prediction output:

$$F_{\theta}^{(i)}(\{u_G(t - \tau_l)\}_{l=1}^m; G)(h_n) = \sum_{j=1}^r b_j^{(i)} \cdot y_j^{(n)}. \quad (9)$$

In Eq. (9), $\{u_G(t - \tau_l)\}_{l=1}^m$ denotes the historical graph traffic states used by the Branch network, τ_l is the l th time lag, h_n is the future temporal query, r denotes the latent basis dimension of the branch-trunk fusion, $b_j^{(i)}$ is the j th basis value generated by the Branch network for node or sample i , and $y_j^{(n)}$ is the corresponding coefficient generated by the Trunk network for the query horizon h_n .

The loss function is defined as the mean absolute error between the observed values $u_G^{(n)}(t + h_n) \in \mathbb{R}^{d \times N \times 1}$ and the predicted values $F_{\theta,v}(\{u_G(t - \tau_l)\}_{l=1}^m; G)(h_n) \in \mathbb{R}^{d \times N \times 1}$, as shown in Eq. (10):

$$L(\theta) = \frac{1}{N} \sum_{v=1}^N |u_{G,v}^{(n)}(t + h_n) - F_{\theta,v}(\{u_G(t - \tau_l)\}_{l=1}^m; G)(h_n)|_1. \quad (10)$$

4.2.1. Branch network

The Branch Network is composed of stacked ST-Blocks, which handle historical graph-valued state information. The input consists of graph state information $\{u_G(t - \tau_l)\}_{l=1}^m$, where $u_G(t - \tau_l) \in \mathbb{R}^{|S|}$. The output is graph-based basis function values $b \in \mathbb{R}^{r \times |V|}$.

(1) Graph Attention Layer for Extracting Spatial Features.

A traffic network can be viewed as a complex graph structure, where intersections are considered as nodes, and the various road segments are considered as edges. To predict traffic speed, it is essential to consider the traffic speeds of the upstream and downstream segments directly related to the target road segment, as well as the connectivity between nodes.

To better capture the dynamic spatial correlations in the traffic network graph, we design a stacked multi-head Graph Attention Network (GAT) method. This method is used to adaptively capture the correlations between the target road segment or node and other adjacent road segments or nodes. As illustrated in Fig. 2, the graph attention layer is well-suited for inductive learning tasks as it does not rely on a fixed adjacency matrix and can dynamically learn the correlations between nodes or road segments, making it more suitable for learning from graph structures. The implementation of GAT follows [23].

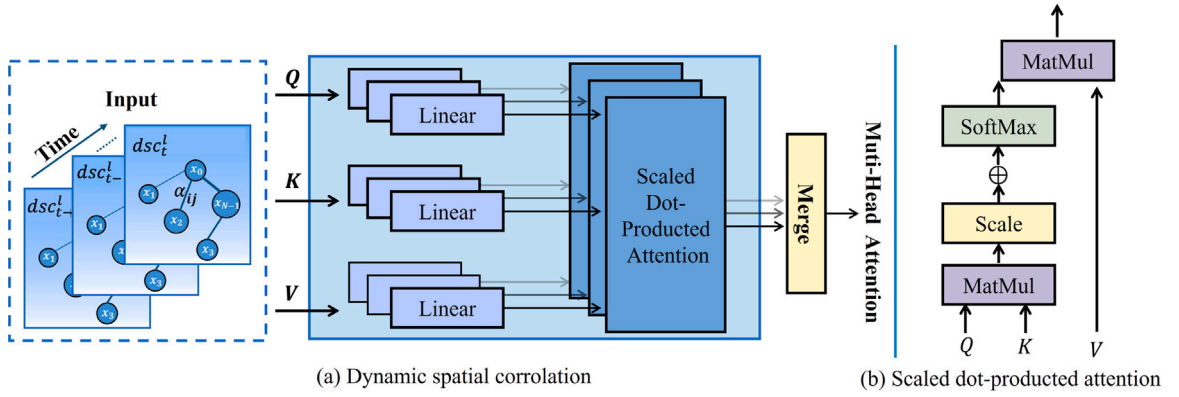


Fig. 2. The structure of graph attention layer.

First, the correlation between nodes is calculated by computing the inner product of the query vector of node v_i and the key vector of node v_j , yielding the correlation $e_{(v_i, v_j)}^{(l, m)}$ between nodes v_i and v_j :

$$e_{(v_i, v_j)}^{(l, m)} = \frac{\langle \sigma_q^m(dsc_{(v_i, t)}^{(l-1)}), \sigma_k^m(dsc_{(v_j, t)}^{(l-1)}) \rangle}{\sqrt{d_e}}, \quad (11)$$

where σ_q^m and σ_k^m are the nonlinear transformation functions used to generate the query and key representations in the m th attention head, respectively, $\langle \cdot, \cdot \rangle$ denotes the inner product operator, and d_e represents the dimension of the node embedding.

Subsequently, to calculate the spatial attention coefficient $\alpha_{(v_i, v_j)}^{(l, m)}$ at time t for the l th layer between nodes v_i and v_j , we use Eq. (12):

$$\alpha_{(v_i, v_j)}^{(l, m)} = \text{softmax}\left(e_{(v_i, v_j)}^{(l, m)}\right) = \frac{\exp\left(e_{(v_i, v_j)}^{(l, m)}\right)}{\sum_{v_r \in \mathcal{V}} \exp\left(e_{(v_i, v_r)}^{(l, m)}\right)}. \quad (12)$$

After obtaining the spatial attention coefficient $\alpha_{(v_i, v_j)}^{(l, m)}$ at time t for the l th layer, the dynamic spatial correlation $dsc_{(v_i, t)}^l$ at time t for the l th layer is calculated using Eq. (13):

$$dsc_{(v_i, t)}^l = \sum_{v_r \in \mathcal{V}} \alpha_{(v_i, v_r)}^{(l, m)} \sigma_v^m(dsc_{(v_r, t)}^{(l-1)}), \quad (13)$$

where σ_v^m denotes the nonlinear transformation for values in the m th attention head.

Finally, the overall dynamic spatial correlation $dsc_{(v_i, t)}$ can be calculated by combining the outputs from multiple heads, as shown in Eq. (14):

$$dsc_{(v_i, t)} = BN\left(\left\| \sum_{m=1}^M dsc_{(v_i, t)}^{(l, m)} W_{ds} + dsc_{(v_i, t)}^{(l-1)} \right\| \right), \quad (14)$$

where BN denotes batch normalization, W_{ds} is a trainable weight matrix, and $\|$ denotes the concatenation operation over multiple attention heads.

(2) Gated Temporal CNNs for Extracting Temporal Features.

Another key issue in traffic flow prediction is capturing the temporal correlations in traffic data. Recurrent Neural Networks (RNNs) are widely used for handling time series data [16]. However, this method is prone to gradient vanishing or exploding, and it has a high computational complexity. As shown in Fig. 3, we designed a Gated Temporal Convolutional Network (Gated TCN), which is employed to capture the temporal dependencies in traffic data. The Gated Temporal Convolutional Network incorporates several temporal convolutional layers with Gated Linear Units (GLUs). Each GLU layer uses different dilation rates, which increase layer by layer, allowing the model to capture temporal features at various scales. This structure ensures that certain features are selectively passed through the gating mechanism, while important information from the original input is preserved through residual connections.

Given input data $X \in \mathbb{R}^{N \times T \times d}$, the data is processed through the gated temporal convolution mechanism as shown below. The GLU mechanism [25] is crucial for regulating the information flow through the network and can be mathematically represented as Eq. (15):

$$GLU(X) = \sigma_g(X * W_g + a) \odot \tanh(X * W_f + b), \quad (15)$$

where $\sigma_g(\cdot)$ denotes the sigmoid activation function used in the gating branch, $\tanh(\cdot)$ denotes the hyperbolic tangent activation function, W_g and W_f are trainable convolutional weight matrices for the gate and feature branches, respectively, and a, b are the corresponding bias terms. The operator $\odot$ represents element-wise multiplication, and $*$ denotes convolution.

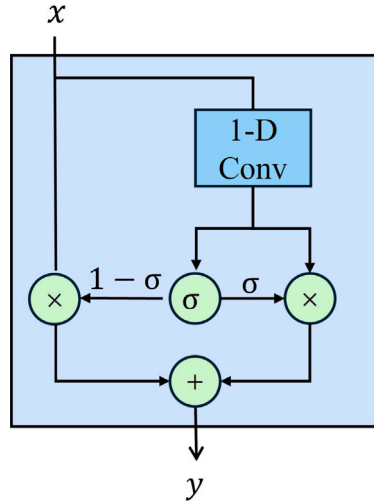


Fig. 3. The gated temporal convolution mechanism.

The Gated Temporal Convolutional Network can then extract the temporal correlation of the data as shown in Eq. (16):

$$dte(X) = GLU(X) \odot (X \times W_I + c) + (1 - GLU(X)) \odot (X \times W_U + d), \quad (16)$$

where X is the input tensor, W_I and W_U are trainable weight matrices, and c, d are bias terms.

(3) S-T Block Fusion Mechanism. Through the spatial attention mechanism and the gated convolution operation, we respectively obtain the dynamic spatial correlation $dsc \in \mathbb{R}^{k \times N \times d}$ and temporal correlation $dte \in \mathbb{R}^{k \times N \times d}$ of the traffic network. Subsequently, we fuse the correlations by employing the gated fusion mechanism to acquire the dynamic spatio-temporal correlation of the traffic network. The formulation is given in Eq. (17) and Eq. (18):

$$dstc = F_s(S, T) \odot dsc + (1 - F_s(S, T)) \odot dte, \quad (17)$$

$$F_s(S, T) = \sigma_g(dsc \odot dte), \quad (18)$$

where σ_g denotes the sigmoid activation function used for gated fusion and $F_s(S, T) \in \mathbb{R}^{k \times N \times d}$ is responsible for regulating the proportion of information flowing through the network.

4.2.2. Trunk network

The Trunk Network employs an MLP architecture to generate trainable coefficients that depend solely on the future temporal query. In contrast to the Branch network, which focuses on extracting spatio-temporal correlations from historical traffic data, the Trunk network is responsible for encoding the query information in the output function domain. Specifically, the Trunk network takes the prediction timestamps as its input and maps them into a continuous latent space. It is implemented as a multi-layer perceptron (MLP) consisting of several fully connected layers with nonlinear activation functions. Through this design, the Trunk network generates a set of temporal feature vectors that represent the functional dependence of the operator on the prediction horizon. These vectors serve as the basis functions of the output domain. The outputs of the Trunk network are subsequently fused with the Branch network outputs through a dot product operation. This allows us to predict the solution operator F at any given time $h_n \in [0, h]$. In this framework, the Trunk network plays a crucial role in linking the prediction horizon with the learned operator representation, thereby enabling DSTGAON to produce accurate predictions at query-specified future time steps.

5. Results and discussion

5.1. Dataset description and processing

5.1.1. Dataset description

We validate the designed model on four real-world datasets: METR-LA, PEMS-BAY, UTD-19 (London), and NYC-Taxi. Each dataset includes traffic data, their corresponding geographic information attributes, and the timestamps of data collection.

METR-LA [19]: This dataset contains traffic speed data from the Los Angeles metropolitan area, collected from March 1, 2012, to June 30, 2012. The data is gathered by 207 loop detectors on the Los Angeles Road network, with a collection frequency of 5 min. The total number of observed traffic data points is 7,094,304. This dataset is considered particularly challenging for traffic prediction tasks due to the significant amount of missing data. There are 575,302 missing data points, accounting for 8.07% of the

total data volume. Moreover, there are instances of prolonged consecutive data gaps. For example, all detectors lack data from 8:25 AM on March 26, 2012, to 4:05 PM on March 27, 2012. Additionally, detector 717490 is missing all data from 12:25 PM on June 5, 2012, to 8:35 AM on June 11, 2012. Consequently, prediction accuracy on this dataset tends to be relatively lower.

PEMS-BAY [19]: This dataset contains traffic speed data from the Bay Area highways, collected between January 1, 2017, and May 31, 2017. The data is gathered by 325 sensors deployed by the California Transportation Agencies (CalTrans) Performance Measurement System (PeMS), with a collection frequency of 5 min. The total number of observed traffic data points is 16,937,700. This dataset demonstrates excellent completeness, with only 521 missing data points. As a result, prediction accuracy for this dataset is expected to be higher, leading to more favorable experimental results.

UTD-19 (London) [40]: The UTD19 dataset, released by ETH Zurich, is a large-scale, multi-city traffic detector dataset covering 40 cities worldwide. It includes traffic flow, speed, and occupancy measurements from over 23,000 fixed road detectors, spanning approximately 2017–2019. Each detector is geocoded and can be aligned with the corresponding road network. UTD19 provides high-quality real-world data for studying cross-city traffic patterns, macroscopic traffic dynamics, and complex spatio-temporal dependencies. In this study, we specifically utilize the London city data with a temporal resolution of 15 min and select the 200 sensors with the highest coverage for analysis.

NYC-Taxi [32]: The New York City Taxi Trip Record Data, provided by the New York City Taxi and Limousine Commission (TLC), includes trip records for yellow taxis, green taxis, and for-hire vehicles (such as Uber and Lyft), covering pickup and drop-off times, locations or zone IDs, trip distances, fares, payment types, and passenger counts. We utilize the preprocessed data from UniST, which covers taxi flows across 192 grid zones, and can be used for analyzing urban traffic patterns, spatio-temporal travel behavior, and travel demand dynamics.

5.1.2. Data processing

To ensure the consistency of the experiment, we use the same method to preprocess the dataset. We use linear interpolation to fill in the missing values in the datasets and apply Z-Score normalization to standardize the data in both datasets. The road traffic network's adjacency matrix is constructed using a Gaussian threshold kernel [41] based on the distance between nodes. Thus, the weighted adjacency matrix A can be expressed as Eq. (19):

$$a_{ij} = \begin{cases} \exp\left(-\frac{\text{dist}(v_i, v_j)^2}{\sigma_d^2}\right), & \text{if } \exp\left(-\frac{\text{dist}(v_i, v_j)^2}{\sigma_d^2}\right) \geq \epsilon, \\ 0, & \text{otherwise,} \end{cases} \quad (19)$$

where v_i and v_j represent nodes i and j , $\text{dist}(v_i, v_j)$ represents the distance between nodes i and j , σ_d is the standard deviation of pairwise node distances used in the Gaussian kernel, and ϵ is the Gaussian threshold. It is worth noting that, in order to reduce the impact of padding values on model training, samples with more than five consecutive missing data points were discarded. During the testing phase, evaluation metrics are calculated only for non-missing regions to ensure the validity of the results.

5.2. Baseline models and metrics

We compare the predictive performance of the proposed model with the following baselines:

Statistical Serial Regression: ARIMA [8]. Traditional Machine Learning: **SVR [9].** Convolutional Neural Network: **LSTNET [42].**

MTGNN: A multivariate time series prediction model based on GNN, which proposes a Mix-hop Propagation layer and a Dilated inception layer to capture complex spatio-temporal dependencies [43].

DCRNN: A spatio-temporal prediction model integrating diffusion convolution and recurrent neural networks, modeling traffic flow dynamics as a diffusion process [19].

ASTGCN: A model combining Graph Convolutional Networks (GCNs) and attention mechanisms, developing spatio-temporal attentional mechanisms to learn dynamic correlations [24].

AGCRN: A model that combines attention, gating, and convolution, with adaptive augmented graph convolution modules and recurrent networks [20].

STGCN: One of the first models using graph neural networks for spatio-temporal data, applying two spectral graph convolution and gated-TCN modules [44].

DeepGraphONet: A deep graph operator network that fuses graph neural networks with deep operator networks to approximate the solution operator of dynamical systems. However, it lacks dedicated mechanisms for modeling non-Markovian memory effects and multi-scale spatio-temporal coupling [45].

iTransformer: An inverted Transformer architecture that embeds entire variate time series as tokens and applies attention and feed-forward networks on the inverted dimensions, achieving strong performance for long-horizon time series forecasting [27].

To evaluate the performance of our model and the baselines, we employ the following three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). The calculations for these metrics are as follows:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (20)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (21)$$

$$MAPE = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad (22)$$

where y_i denotes the true values, $\hat{y}_i$ denotes the predicted values, and N is the number of samples.

5.3. Experiment settings

The hyperparameters of the model mainly include the learning rate, batch size, epochs, early stopping patience, number of ST-blocks, number of hidden unit channels, and the number of heads in multi-head attention. During the training process, we consistently use the Adam optimizer with an initial learning rate of 0.005. The batch size is set to 128, and the number of training epochs is 1000. To prevent overfitting during training, we introduce an early stopping strategy with an early stopping patience set to 100 epochs. The number of ST-blocks, the number of hidden unit channels within these blocks, and the number of heads in multi-head attention are crucial hyperparameters that significantly impact the model's performance. Different combinations of these hyperparameters can lead to substantial variations in predictive accuracy. To select the optimal hyperparameter values, we employ a grid search method, evaluating the performance of each combination on the validation set to identify the best parameter configuration. All experiments utilize the past 60 min of historical data to predict traffic conditions for the next 15, 30, and 60 min. The proposed model and baseline models are implemented in the PyTorch framework, and all experiments are conducted using an Intel(R) i7-12700F CPU @ 2.10 GHz and an NVIDIA GeForce RTX 3070 GPU. We divide the datasets such that 70% of the data is used for training, 20% for testing, and the remaining 10% for validation.

In our experiments, we conduct pre-experiments with an exhaustive grid search over all possible hyperparameter combinations. We select the number of ST-Blocks from $\{1, 2, 3, 4, 5\}$, the number of hidden unit channels from $\{8, 16, 32, 64, 128\}$, and the number of heads in multi-head attention from $\{1, 2, \dots, 10\}$. The final Branch network adopts 4 ST-Blocks, each with 128 hidden unit channels, and 6 attention heads in the attention mechanism layer. Each ST-Block contains a temporal block with 3 gated convolutional layers, and the graph attention layer uses 6 attention heads. Dropout, batch normalization, and residual connections are integrated within each block. These choices are selected based on loss and validation metric performance, effectively balancing long-horizon prediction accuracy with computational efficiency. For each baseline model, we adopt the optimal parameter configurations proposed in their respective papers and retrain them on the adopted datasets to ensure fairness.

5.4. Experiment results and analysis

5.4.1. Prediction performance comparison

Predicting highway traffic flow remains a challenging task due to the nonlinear and complex spatio-temporal dependencies in traffic systems. A key difficulty is that prediction errors tend to accumulate and propagate over time, leading to degraded accuracy at longer horizons. To evaluate the effectiveness of the proposed model, we conducted comparative experiments against several widely used baseline models.

Table 2 presents the performance of the DSTGAON model and other baselines for 15-, 30-, and 60-minute prediction tasks on the METR-LA and PEMS-BAY datasets. The analysis below focuses on the experimental results from the METR-LA dataset as an example. Traditional models such as ARIMA and SVR exhibit the weakest performance across all horizons. For instance, at the 15-minute horizon, their MAE values (4.81 and 4.76) are substantially higher than those of deep learning methods. This gap highlights the limited ability of conventional statistical and machine learning models to capture nonlinear and complex spatio-temporal correlations in traffic data. Early deep learning methods such as LSTNet improve short-term predictions by leveraging CNN and RNN modules to extract local temporal features. At 15 min, LSTNet reduces MAE by 19.3% compared with ARIMA and by 14.3% compared with SVR. However, without explicit modeling of spatial dependencies, its performance quickly deteriorates with longer horizons.

Graph-based deep models show significant superiority by effectively capturing spatial correlations. For example, at the 15-minute horizon, models such as MTGNN, DCRNN, ASTGCN, AGCRN, and STGCN, reduce MAE by 30–40% relative to LSTNet. These improvements confirm that exploiting the irregular graph structure of road networks is crucial for accurate traffic prediction. Our proposed DSTGAON consistently achieves the best performance across all horizons. At 15 min, DSTGAON achieves an RMSE of 4.14, which is 13–18% lower than those of above baseline. At 60 min, the advantage becomes even more pronounced: the RMSE of DSTGAON (5.06) is 19–33% lower than that of the baselines. Similar trends are observed for MAE and MAPE. These results demonstrate that DSTGAON not only provides strong short-term prediction accuracy but also excels at long-term forecasting (30–60 minutes), where error accumulation typically undermines other models.

Compared with DeepGraphONet, DSTGAON further incorporates Mori–Zwanzig-guided memory modeling and gated multi-scale spatio-temporal fusion within ST-Blocks, achieving more effective error suppression at long horizons (e.g., at 60 min on METR-LA, MAE: 2.48 vs. 2.63; RMSE: 5.06 vs. 5.49). In contrast to iTransformer, which models multivariate correlations but lacks explicit spatial graph modeling and operator-level error control, DSTGAON directly approximates the solution operator, yielding substantially lower errors at extended horizons (e.g., at 60 min on METR-LA, MAE: 2.48 vs. 3.40; RMSE: 5.06 vs. 7.19).

To further assess the generalization capability of DSTGAON, we conduct additional experiments on two datasets of different traffic data types: UTD-19 (London), which provides fixed detector measurements similar to METR-LA and PEMS-BAY, and NYC-Taxi, which captures mobility flows from taxi trip records aggregated into grid zones. Table 3 reports the prediction performance of all models at the 60-minute horizon on these two datasets. Across both datasets, DSTGAON consistently achieves the lowest errors across both datasets. On UTD-19, it outperforms the strongest baselines DeepGraphONet and iTransformer across all metrics.

Table 2

Model performance comparison on METR-LA and PEMS-BAY datasets.

Dataset	Model	$T = 15$ min			$T = 30$ min			$T = 60$ min		
		MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
METR-LA	ARIMA	4.81	10.02	12.15	4.94	12.84	14.52	4.98	12.67	15.17
	SVR	4.76	8.21	9.60	5.15	10.45	12.70	6.90	13.23	17.40
	LSTNet	4.08	7.00	8.22	4.37	7.12	9.23	4.78	9.22	12.56
	MTGNN	2.54	4.80	6.31	2.78	5.58	7.43	3.14	6.55	9.00
	DCRNN	2.61	5.01	6.90	2.99	5.94	8.33	3.68	7.50	9.71
	ASTGCN	2.59	5.06	6.85	2.93	5.93	8.11	3.41	7.03	9.90
	AGCRN	2.52	4.76	6.39	2.76	5.49	7.38	3.10	6.36	8.61
	STGCN	2.84	4.92	7.22	3.09	5.50	8.07	3.45	6.23	9.32
	DeepGraphONet	2.23	4.24	5.38	2.41	4.65	6.01	2.63	5.49	6.77
	iTransformer	2.65	5.11	6.85	2.97	6.00	8.13	3.40	7.19	9.39
	Ours	2.16	4.14	5.27	2.33	4.47	5.87	2.48	5.06	6.37
PEMS-BAY	ARIMA	3.33	6.69	8.10	3.33	6.69	8.10	3.33	6.68	8.10
	SVR	1.62	3.30	3.50	2.33	4.76	5.40	3.38	6.50	8.30
	LSTNet	1.64	3.22	3.47	2.38	4.38	5.04	2.97	5.52	6.86
	MTGNN	1.24	2.53	2.57	1.46	3.22	3.14	1.80	4.13	4.12
	DCRNN	1.24	2.57	2.61	1.54	3.43	3.47	2.07	4.90	5.26
	ASTGCN	1.32	2.72	2.80	1.56	3.37	3.48	1.87	4.14	4.30
	AGCRN	1.24	2.48	2.59	1.44	3.14	3.15	1.73	3.87	3.96
	STGCN	1.41	2.65	2.98	1.60	3.18	3.49	1.83	3.89	4.12
	DeepGraphONet	1.25	2.49	2.61	1.51	3.15	3.17	1.71	3.80	3.78
	iTransformer	1.32	2.62	2.80	1.63	3.68	3.63	1.92	4.59	4.21
	Ours	1.22	2.47	2.57	1.42	3.10	3.05	1.66	3.82	3.65

Table 3Model performance comparison on UTD-19 (London) and NYC-Taxi datasets at $T = 60$ min.

Model	UTD-19 (London)			NYC-Taxi		
	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
AGCRN	92.06	189.76	19.05	41.44	84.52	60.55
ASTGCN	91.99	211.86	19.04	21.66	50.05	31.64
DCRNN	88.05	184.05	18.22	21.55	40.85	31.48
MTGNN	87.27	178.66	18.06	20.27	47.31	29.61
STGCN	86.96	190.53	18.00	20.01	44.19	29.23
iTransformer	81.01	141.47	16.77	18.78	40.34	27.40
DeepGraphONet	80.37	157.05	16.63	11.61	25.79	16.97
Ours	79.58	150.50	16.47	10.85	24.42	15.85

Table 4Component ablation analysis of DSTGAON on METR-LA at $T = 60$ min.

	Component			Metric		
	Operator framework	Graph attention	Gated TCN	MAE	RMSE	MAPE (%)
Ablation 1	–	✓	–	3.57	7.31	8.44
Ablation 2	–	✓	✓	3.34	6.96	7.40
Ablation 3	✓	✓	–	3.09	6.98	7.09
Ours	✓	✓	✓	2.48	5.06	6.37

On NYC-Taxi, which represents a fundamentally different data modality (mobility flows rather than sensor measurements), the relative advantage of DSTGAON is even larger, demonstrating its strong generalization capability across diverse traffic data types and network structures. The superior performance of DSTGAON can be attributed to its operator learning framework combined with spatio-temporal blocks and multi-head attention, which jointly capture both nonlinear temporal dependencies and graph-based spatial correlations.

5.4.2. Ablation experiments

To assess the contribution of each key component in DSTGAON, we conduct a component-level ablation study on the METR-LA dataset at the 60-minute horizon. We examine three core designs: the branch–trunk operator framework (Operator Framework), the graph attention mechanism (Graph Attention), and the gated temporal convolution network (Gated TCN). By selectively enabling or disabling each component, we obtain three model variants whose performance is summarized in [Table 4](#).

Starting from Ablation 1 (Graph Attention only), adding Gated TCN (Ablation 2) reduces MAE from 3.57 to 3.34 and MAPE from 8.44% to 7.40%, confirming that the gating mechanism effectively captures multi-scale temporal features. Alternatively, adding the

Table 5
Performance of different models with and without the operator framework.

Model	$T = 15 \text{ min}$			$T = 30 \text{ min}$			$T = 60 \text{ min}$		
	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
DCRNN	2.61	5.01	6.90	2.99	5.94	8.33	3.68	7.50	9.71
+ Operator Framework	2.37	4.59	5.87	2.62	5.30	6.80	2.96	6.17	8.12
STGCN	2.84	4.92	7.22	3.09	5.50	8.07	3.45	6.23	9.32
+ Operator Framework	2.22	4.17	5.34	2.41	4.60	5.81	2.59	5.18	6.53
ASTGCN	2.59	5.06	6.85	2.93	5.93	8.11	3.41	7.03	9.90
+ Operator Framework	2.26	4.62	6.07	2.47	5.67	6.75	2.98	6.48	8.35
MTGNN	2.54	4.80	6.31	2.78	5.58	7.43	3.14	6.55	9.00
+ Operator Framework	2.20	4.19	5.33	2.33	4.63	5.90	2.52	5.19	6.72
AGCRN	2.52	4.76	6.39	2.76	5.49	7.38	3.10	6.36	8.61
+ Operator Framework	2.29	4.34	5.72	2.42	4.76	6.26	2.57	5.23	6.93

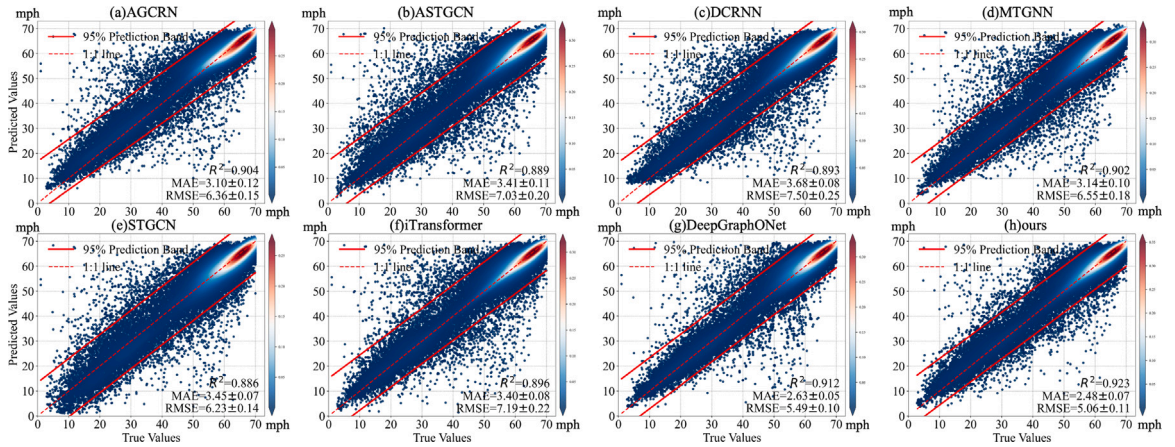


Fig. 4. Comparison of fitting performance (mph is miles per hour).

Operator Framework (Ablation 3) reduces MAE more substantially from 3.57 to 3.09 and MAPE from 8.44% to 7.09%, indicating that the branch–trunk architecture contributes more to absolute error reduction than the temporal gating alone. Finally, integrating all three components (Ours) yields the best results across all metrics, with MAE reaching 2.48 (reduced by 19.7%–30.5% relative to the three ablations) and RMSE reaching 5.06 (reduced by 27.1%–30.8%). These results confirm that each component provides complementary benefits and that their combination is essential for achieving optimal long-horizon prediction performance.

To further validate the effectiveness and generalizability of the proposed operator learning framework, we conduct experiments by integrating the branch–trunk architecture into five representative baseline models: DCRNN, STGCN, ASTGCN, MTGNN, and AGCRN. Importantly, we do not modify the internal architectures or hyperparameters of the original models. Instead, we use these original models to fully replace the spatio-temporal modeling in the branch network, while the trunk network remains responsible for handling temporal queries. Table 5 presents the results. Across all five baselines, embedding the operator framework consistently yields substantial improvements, with MAE reductions ranging from 9.1% to 21.8% at 15 min, 12.3% to 22.0% at 30 min, and 12.6% to 24.9% at 60 min. In most cases, the gains become larger as the prediction horizon extends, suggesting that the operator framework is particularly beneficial for long-horizon forecasting. Furthermore, since the framework requires no modification to the original models' architectures or hyperparameters, these results demonstrate its strong universality and plug-and-play capability across diverse spatio-temporal prediction models.

In summary, the operator framework not only strengthens our proposed DSTGAON model but also serves as a general enhancement module that can be directly embedded into diverse traffic forecasting architectures. This demonstrates its strong potential for wide adoption in real-world intelligent transportation systems.

5.4.3. Fitting performance

To further validate the superiority of the proposed model, we compare its 60-minute prediction results with those of strong baseline models. The fitted results are visualized against ground-truth values in Fig. 4, leading to the following observations:

We assess the models using R^2 , MAE, and RMSE. Here, R^2 denotes the coefficient of determination, which measures the goodness of fit between predicted and observed values. Among all models, Ours achieves the highest R^2 (0.923), along with the lowest MAE (2.48) and RMSE (5.06), indicating the smallest errors and the most accurate predictions. The scatter plots in Fig. 4 illustrate the degree of agreement between predicted and actual values. Compared with the baselines, the predictions of DSTGAON are more

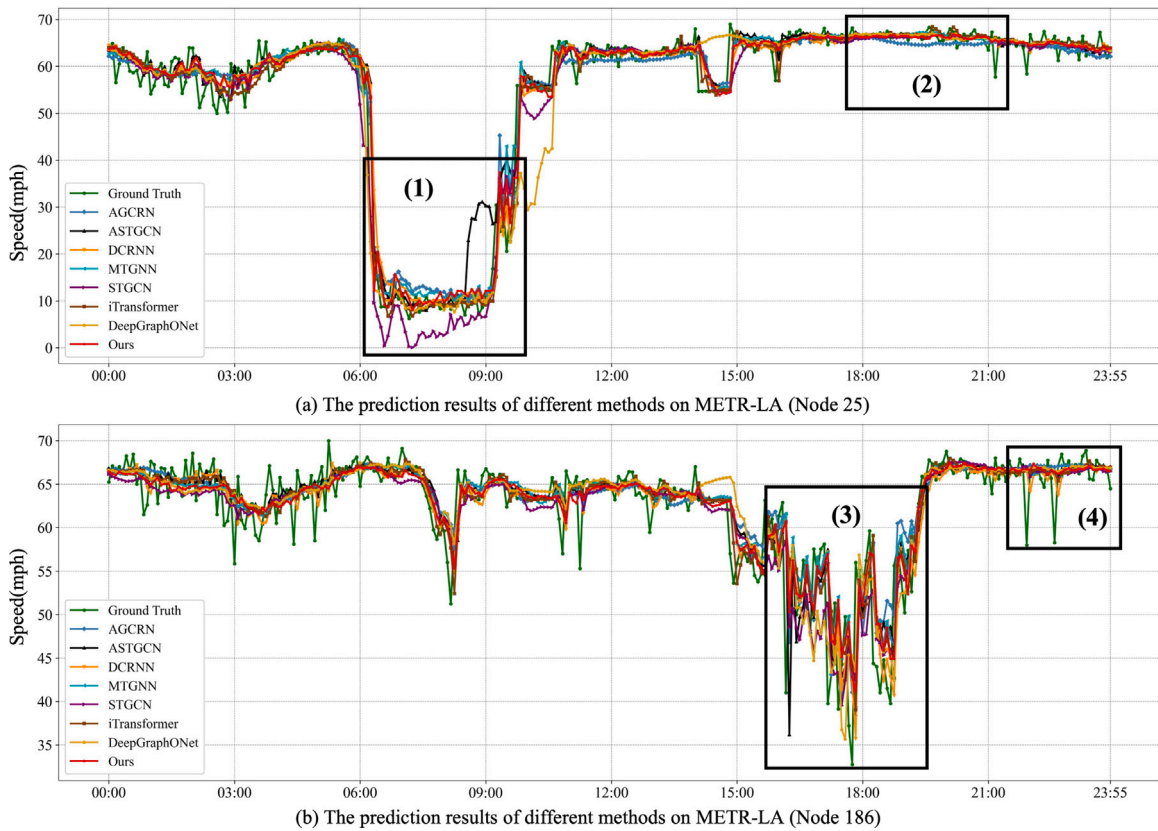


Fig. 5. The prediction results. (a) The prediction results of different methods on METR-LA (Node 186). (b) The prediction results of different methods on METR-LA (Node 25).

tightly concentrated around the diagonal, with noticeably fewer points falling outside the 95% confidence interval and the narrowest interval width among all models. This visually demonstrates the superior stability and predictive reliability of the proposed model.

To further quantify the statistical significance of the improvements, we trained the strongest baseline model and DSTGAON under five different random seeds and report their $MAE \pm std$ in Fig. 4. A paired t -test confirms that the improvement of DSTGAON over the best baseline is statistically significant ($p < 0.01$).

5.4.4. Stability analysis of the proposed model

To evaluate the stability of the proposed model, we select two representative nodes from the METR-LA dataset: Node 186, which exhibits significant morning peak characteristics, and Node 25, which exhibits significant evening peak characteristics. Fig. 5 visualizes the prediction results of different models at these two nodes. Our model not only detects the start and end times of peak hours but also captures the complex temporal variations in traffic flow, generating prediction curves that closely match the ground truth. This indicates its strong ability to accurately reproduce the overall traffic speed trends.

For a more detailed evaluation, we analyze the prediction errors across different time periods, as highlighted by the four rectangular regions in Fig. 5. The absolute difference between predicted and actual values is defined as the prediction error, and both the mean and standard deviation (std) of errors are calculated for each model. The results in Fig. 6 show that our model achieves the lowest prediction errors across both peak and off-peak periods. The mean errors are 1.32 and 1.16 for Nodes 186 and 25 during peak periods, and 0.48 and 0.71 during off-peak periods, respectively, all of which are lower than those of the competing models. A similar pattern is observed for the standard deviation of errors, where our model again records the smallest values, with 1.64 and 0.77 during peak periods and 0.42 and 0.71 during off-peak periods.

These results confirm that the proposed model not only provides higher accuracy but also demonstrates superior stability and robustness in capturing traffic speed patterns under varying demand conditions. Overall, these findings collectively confirm that our proposed model achieves the highest accuracy and stability among all baselines. More importantly, it maintains consistent predictive performance across both peak and off-peak periods, demonstrating its robustness for real-world traffic speed forecasting tasks where demand patterns vary significantly over time.

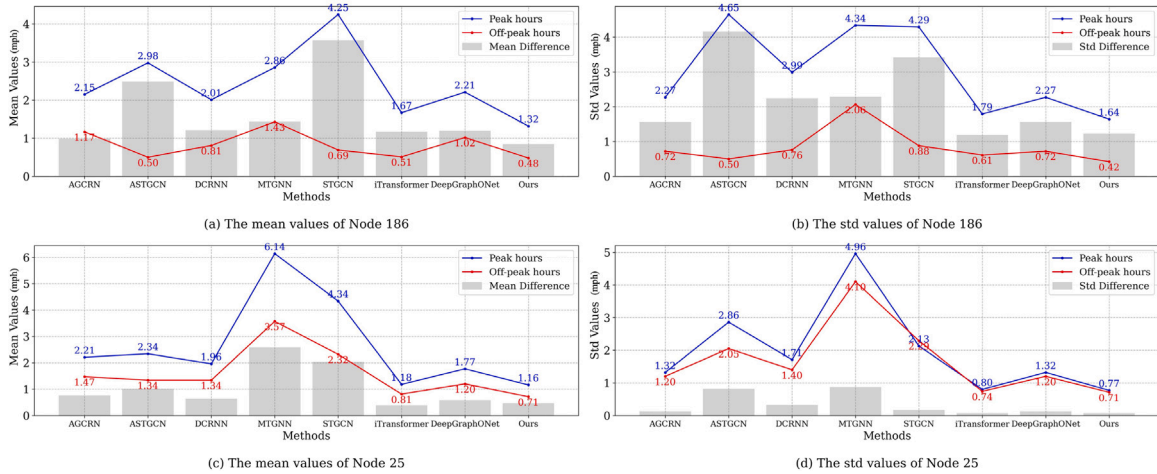


Fig. 6. The mean and standard deviation for different methods.

5.4.5. Computational efficiency analysis

We evaluate the computational complexity of DSTGAON in terms of parameter count, inference speed, and GPU memory usage. All measurements are conducted on an NVIDIA GeForce RTX 3070 GPU. On METR-LA and PEMS-BAY, DSTGAON demonstrates a moderate parameter size and memory footprint, with inference times of 16.74 ms and 28.26 ms per batch, respectively, both well below 1 s. These results indicate that the model is suitable for real-time traffic prediction even on large-scale networks.

6. Conclusion

In this paper, we propose DSTGAON, a novel operator learning framework for traffic prediction that integrates a branch net and trunk net to approximate nonlinear dynamic operators over spatio-temporal graphs. Grounded in the Mori–Zwanzig formalism, the model explicitly captures non-Markovian memory effects through discrete historical state approximation, while the gated multi-scale spatio-temporal fusion within ST-Blocks enables effective modeling of complex traffic dynamics. Extensive experiments on four real-world datasets (METR-LA, PEMS-BAY, UTD-19, and NYC-Taxi) spanning different traffic data modalities demonstrated the superiority of the proposed approach. DSTGAON consistently achieved the lowest MAE, RMSE, and MAPE across multiple prediction horizons, with its advantage becoming more pronounced at longer horizons (30–60 min) and on datasets of different data types. Stability analysis confirmed robust performance during both peak and off-peak periods. Moreover, embedding the operator framework into existing models such as DCRNN, STGCN, ASTGCN, MTGNN, and AGCRN yielded consistent accuracy gains, highlighting its universality and plug-and-play capability. Overall, this study establishes a theoretical and practical foundation for applying operator learning in intelligent transportation systems. For future work, we aim to explore the generalization performance of the proposed framework, particularly by transferring trained models to unseen traffic networks, and to further enhance the attention mechanisms for mining complex spatio-temporal dependencies under diverse traffic conditions.

CRedit authorship contribution statement

Hao Tong: Writing – review & editing, Validation, Methodology, Conceptualization. **Qi Ai:** Writing – original draft, Visualization, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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